Aditya Mittha

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Summary

Final-year Electronics and Telecommunication engineering student with hands-on experience building embedded systems around ESP32 and ARM-based microcontrollers. Proficient in Embedded C and Python with practical knowledge of communication protocols (UART, I2C, CAN, MQTT), interrupt-driven programming, and FreeRTOS task scheduling. Experienced in IoT product development — from sensor firmware to AWS IoT Core cloud integration. Comfortable debugging with logic analyzers, working on Linux environments, and collaborating with cross-functional teams.

Skills

Programming Languages: Embedded C, Python, Verilog, Bash/Shell Scripting

Microcontrollers & Hardware: ESP32, ARM-based MCUs, Raspberry Pi, Arduino, Logic Analyzer

Protocols & Interfaces: UART, SPI, I2C, CAN, MQTT (QoS), GPIO, ADC, Interrupts (ISR)

Embedded & RTOS: FreeRTOS, Linux, Timers/Counters, Watchdog Timer, Low Power Modes, SWD/JTAG Debugging

Cloud & IoT Platforms: AWS IoT Core, Firebase, MQTT Broker (Mosquitto), Git, GitHub

Data & Analytics: Machine Learning (scikit-learn), Data Structures, NumPy, pandas

Soft Skills: Problem Solving, Team Collaboration, Self-Learning

Education

Bachelor's Degree in Electronics and Telecommunication 2027
Walchand Institute Of Technology *CGPA: 9.27*

Higher Schooling 2023
D.H.B Soni College, Solapur *Percentage: 83%*

Work Experience

Embedded System Intern May 2025 – June 2025
Codec Technologies India *Remote*

- Configured and tested UART and I^2C communication interfaces on microcontrollers, writing Embedded C code to establish reliable data transmission between modules.
- Identified and resolved 10+ hardware interfacing defects across two embedded modules before production testing.
- Managed version control with Git/GitHub across 2 active branches, maintaining project documentation and development workflows.

Technologies: Embedded C, UART, I^2C, GPIO, Git, GitHub

Projects

AQUANOVA – Smart Water Pressure Management System 2026

- Wrote ESP32 firmware to poll pressure and flow sensors over ADC/GPIO, publishing readings via MQTT (QoS) to Firebase cloud storage for real-time pipeline monitoring.
- Built a Python analytics pipeline with pandas, numpy, and scikit-learn's Isolation Forest — detected leaks and pressure anomalies, reducing estimated maintenance response time.

Role: IoT Firmware & Cloud Integration (3-person team)

Tech Stack: ESP32, MQTT (QoS), Firebase, ADC, Pressure/Flow Sensors, Python, Telegram/SMS alerts

SORTIFY – Waste Segregation System 2025

- Designed an AI-driven waste sorting system combining computer vision, weight sensing, and vibration-based sorting to classify household waste into 4 recyclable categories.
- Integrated Raspberry Pi with camera module, weight sensor, and servo motor; automated identification, measurement, and bin control using Python and C.
- Built a IoT dashboard tracking real-time segregation statistics, giving municipal teams actionable waste management data.

Role: Embedded Hardware & Raspberry Pi Integration (Hackathon, 4-person team)

Tech Stack: Raspberry Pi, Embedded C, Python, Camera Module, HX711, Servo/PWM, Firebase, IoT Cloud

Rewards & Achievements

- 2x Winner, National Electronic Quiz Competition (Quizotronics)
- Runner-up, State-level chess championship
- Participated in 4^+ Hackathons

Languages

English | Hindi | Marathi | Kannada